

HACKWithInfy

DSA Interview Preparation

25-Day Intensive Roadmap

200 Curated High-Impact Problems · 8 Questions Per Day

25

Days

200

Questions

8

Per Day

Topics Covered

*Arrays · Strings · Linked Lists · Stacks & Queues · Trees · Binary
Search · Hashing · Graphs · Dynamic Programming · Heaps · Tries · Backtracking · Bit
Manipulation*

How to Use This Roadmap

Follow these guidelines to get the most out of this 25-day plan:

1. Solve all 8 problems each day before checking solutions.
2. Time yourself: aim for 20–30 min per problem.
3. After solving, review time & space complexity.
4. Re-attempt any skipped problems the next morning.
5. For every Hard problem: understand the pattern, not just the answer.
6. On Day 25 simulate a real interview: no hints, timed.

Recommended Resources

Platform	Why Use It
LeetCode	Primary platform – search by problem name.
GeeksForGeeks	Great for theory & alternative approaches.
Infosys HackWithInfy portal	Register and take mock tests in Week 3–4.

Difficulty Key

🟢 Easy Foundation problems. Complete all.	🟡 Medium Core interview problems. Focus here.	🔴 Hard Differentiators. Aim for understanding.
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DAY 1 — Arrays – Basics & Two Pointers *Easy*

#	Problem	Status
1	Two Sum	☐ Solved ☐ Review
2	Best Time to Buy and Sell Stock	☐ Solved ☐ Review
3	Majority Element	☐ Solved ☐ Review
4	Move Zeroes	☐ Solved ☐ Review
5	Squares of a Sorted Array	☐ Solved ☐ Review
6	Merge Sorted Array	☐ Solved ☐ Review
7	Remove Duplicates from Sorted Array	☐ Solved ☐ Review
8	Sort Array By Parity	☐ Solved ☐ Review

DAY 2 — Arrays – Sliding Window & Prefix Sum *Easy-Medium*

#	Problem	Status
1	Max Consecutive Ones III	☐ Solved ☐ Review
2	Subarray Sum Equals K	☐ Solved ☐ Review
3	Contiguous Array	☐ Solved ☐ Review
4	Range Sum Query – Immutable	☐ Solved ☐ Review
5	Find Pivot Index	☐ Solved ☐ Review
6	Product of Array Except Self	☐ Solved ☐ Review
7	Maximum Subarray	☐ Solved ☐ Review
8	Running Sum of 1d Array	☐ Solved ☐ Review

DAY 3 — Arrays – Intervals & Sorting *Medium*

#	Problem	Status
1	Merge Intervals	☐ Solved ☐ Review
2	Insert Interval	☐ Solved ☐ Review
3	Non-overlapping Intervals	☐ Solved ☐ Review
4	Interval List Intersections	☐ Solved ☐ Review
5	3Sum	☐ Solved ☐ Review

6	Container With Most Water	☐ Solved ☐ Review
7	Sort Colors	☐ Solved ☐ Review
8	Rotate Array	☐ Solved ☐ Review

DAY 4 — Strings – Fundamentals *Easy*

#	Problem	Status
1	Valid Palindrome	☐ Solved ☐ Review
2	Valid Anagram	☐ Solved ☐ Review
3	First Unique Character in a String	☐ Solved ☐ Review
4	Longest Common Prefix	☐ Solved ☐ Review
5	Is Subsequence	☐ Solved ☐ Review
6	Reverse Words in a String	☐ Solved ☐ Review
7	Isomorphic Strings	☐ Solved ☐ Review
8	String Compression	☐ Solved ☐ Review

DAY 5 — Strings – Sliding Window & Advanced *Medium*

#	Problem	Status
1	Longest Substring Without Repeating Characters	☐ Solved ☐ Review
2	Find All Anagrams in a String	☐ Solved ☐ Review
3	Group Anagrams	☐ Solved ☐ Review
4	Permutation in String	☐ Solved ☐ Review
5	Longest Repeating Character Replacement	☐ Solved ☐ Review
6	Minimum Window Substring	☐ Solved ☐ Review
7	Longest Palindromic Substring	☐ Solved ☐ Review
8	Count and Say	☐ Solved ☐ Review

DAY 6 — Linked Lists *Easy–Medium*

#	Problem	Status
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1	Reverse Linked List	☐ Solved ☐ Review
2	Merge Two Sorted Lists	☐ Solved ☐ Review
3	Linked List Cycle	☐ Solved ☐ Review
4	Remove Nth Node From End of List	☐ Solved ☐ Review
5	Middle of the Linked List	☐ Solved ☐ Review
6	Palindrome Linked List	☐ Solved ☐ Review
7	Add Two Numbers	☐ Solved ☐ Review
8	Copy List with Random Pointer	☐ Solved ☐ Review

DAY 7 — Linked Lists – Advanced *Medium*

#	Problem	Status
1	Reverse Linked List II	☐ Solved ☐ Review
2	Odd Even Linked List	☐ Solved ☐ Review
3	Sort List	☐ Solved ☐ Review
4	Reorder List	☐ Solved ☐ Review
5	Rotate List	☐ Solved ☐ Review
6	Remove Duplicates from Sorted List II	☐ Solved ☐ Review
7	Swap Nodes in Pairs	☐ Solved ☐ Review
8	LRU Cache	☐ Solved ☐ Review

DAY 8 — Stacks & Queues *Easy-Medium*

#	Problem	Status
1	Valid Parentheses	☐ Solved ☐ Review
2	Implement Queue using Stacks	☐ Solved ☐ Review
3	Min Stack	☐ Solved ☐ Review
4	Daily Temperatures	☐ Solved ☐ Review
5	Next Greater Element I	☐ Solved ☐ Review
6	Next Greater Element II	☐ Solved ☐ Review
7	Decode String	☐ Solved ☐ Review

8	Evaluate Reverse Polish Notation	☐ Solved ☐ Review
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DAY 9 — Stacks – Advanced *Medium–Hard*

#	Problem	Status
1	Asteroid Collision	☐ Solved ☐ Review
2	Remove K Digits	☐ Solved ☐ Review
3	Basic Calculator II	☐ Solved ☐ Review
4	Largest Rectangle in Histogram	☐ Solved ☐ Review
5	Trapping Rain Water	☐ Solved ☐ Review
6	132 Pattern	☐ Solved ☐ Review
7	Minimum Remove to Make Valid Parentheses	☐ Solved ☐ Review
8	Remove Duplicate Letters	☐ Solved ☐ Review

DAY 10 — Binary Trees – Traversal & Basics *Easy–Medium*

#	Problem	Status
1	Binary Tree Inorder Traversal	☐ Solved ☐ Review
2	Binary Tree Level Order Traversal	☐ Solved ☐ Review
3	Maximum Depth of Binary Tree	☐ Solved ☐ Review
4	Invert Binary Tree	☐ Solved ☐ Review
5	Symmetric Tree	☐ Solved ☐ Review
6	Diameter of Binary Tree	☐ Solved ☐ Review
7	Path Sum	☐ Solved ☐ Review
8	Merge Two Binary Trees	☐ Solved ☐ Review

DAY 11 — Binary Trees – Intermediate *Medium*

#	Problem	Status
1	Binary Tree Right Side View	☐ Solved ☐ Review
2	Lowest Common Ancestor of a Binary Tree	☐ Solved ☐ Review

3	Construct Binary Tree from Preorder and Inorder Traversal	☐ Solved ☐ Review
4	Path Sum II	☐ Solved ☐ Review
5	Path Sum III	☐ Solved ☐ Review
6	Flatten Binary Tree to Linked List	☐ Solved ☐ Review
7	Binary Tree Maximum Path Sum	☐ Solved ☐ Review
8	Serialize and Deserialize Binary Tree	☐ Solved ☐ Review

DAY 12 — Binary Search Trees *Medium*

#	Problem	Status
1	Validate Binary Search Tree	☐ Solved ☐ Review
2	Kth Smallest Element in a BST	☐ Solved ☐ Review
3	Delete Node in a BST	☐ Solved ☐ Review
4	Lowest Common Ancestor of a BST	☐ Solved ☐ Review
5	Convert Sorted Array to Binary Search Tree	☐ Solved ☐ Review
6	Binary Search Tree Iterator	☐ Solved ☐ Review
7	Recover Binary Search Tree	☐ Solved ☐ Review
8	Balance a Binary Search Tree	☐ Solved ☐ Review

DAY 13 — Binary Search *Easy-Medium*

#	Problem	Status
1	Binary Search	☐ Solved ☐ Review
2	Search Insert Position	☐ Solved ☐ Review
3	First Bad Version	☐ Solved ☐ Review
4	Search in Rotated Sorted Array	☐ Solved ☐ Review
5	Find Minimum in Rotated Sorted Array	☐ Solved ☐ Review
6	Find First and Last Position of Element in Sorted Array	☐ Solved ☐ Review
7	Koko Eating Bananas	☐ Solved ☐ Review
8	Capacity To Ship Packages Within D Days	☐ Solved ☐ Review

DAY 14 — Hashing & Maps *Easy-Medium*

#	Problem	Status
1	Contains Duplicate	☐ Solved ☐ Review
2	Ransom Note	☐ Solved ☐ Review
3	Longest Consecutive Sequence	☐ Solved ☐ Review
4	Insert Delete GetRandom O(1)	☐ Solved ☐ Review
5	Find All Duplicates in an Array	☐ Solved ☐ Review
6	Word Pattern	☐ Solved ☐ Review
7	Top K Frequent Elements	☐ Solved ☐ Review
8	Group Anagrams	☐ Solved ☐ Review

DAY 15 — Graphs – BFS & DFS Basics *Medium*

#	Problem	Status
1	Number of Islands	☐ Solved ☐ Review
2	Clone Graph	☐ Solved ☐ Review
3	Rotting Oranges	☐ Solved ☐ Review
4	01 Matrix	☐ Solved ☐ Review
5	Flood Fill	☐ Solved ☐ Review
6	Find if Path Exists in Graph	☐ Solved ☐ Review
7	Max Area of Island	☐ Solved ☐ Review
8	Surrounded Regions	☐ Solved ☐ Review

DAY 16 — Graphs – Intermediate *Medium*

#	Problem	Status
1	Course Schedule	☐ Solved ☐ Review
2	Course Schedule II	☐ Solved ☐ Review
3	Accounts Merge	☐ Solved ☐ Review
4	Word Search	☐ Solved ☐ Review
5	Number of Provinces	☐ Solved ☐ Review
6	Pacific Atlantic Water Flow	☐ Solved ☐ Review

7	Minimum Height Trees	☐ Solved ☐ Review
8	Redundant Connection	☐ Solved ☐ Review

DAY 17 — Graphs – Advanced & Shortest Path *Medium–Hard*

#	Problem	Status
1	Cheapest Flights Within K Stops	☐ Solved ☐ Review
2	Network Delay Time	☐ Solved ☐ Review
3	All Paths From Source to Target	☐ Solved ☐ Review
4	Word Ladder	☐ Solved ☐ Review
5	Shortest Path in Binary Matrix	☐ Solved ☐ Review
6	Critical Connections in a Network	☐ Solved ☐ Review
7	Evaluate Division	☐ Solved ☐ Review
8	Longest Increasing Path in a Matrix	☐ Solved ☐ Review

DAY 18 — Dynamic Programming – Basics *Easy–Medium*

#	Problem	Status
1	Climbing Stairs	☐ Solved ☐ Review
2	House Robber	☐ Solved ☐ Review
3	House Robber II	☐ Solved ☐ Review
4	Coin Change	☐ Solved ☐ Review
5	Maximum Subarray	☐ Solved ☐ Review
6	Jump Game	☐ Solved ☐ Review
7	Decode Ways	☐ Solved ☐ Review
8	Unique Paths	☐ Solved ☐ Review

DAY 19 — Dynamic Programming – Intermediate *Medium*

#	Problem	Status
1	Longest Increasing Subsequence	☐ Solved ☐ Review
2	Longest Common Subsequence	☐ Solved ☐ Review
3	Coin Change II	☐ Solved ☐ Review
4	Partition Equal Subset Sum	☐ Solved ☐ Review
5	Maximal Square	☐ Solved ☐ Review
6	Jump Game II	☐ Solved ☐ Review
7	Target Sum	☐ Solved ☐ Review
8	Word Break	☐ Solved ☐ Review

DAY 20 — Dynamic Programming – Advanced *Medium–Hard*

#	Problem	Status
1	Longest Palindromic Subsequence	☐ Solved ☐ Review
2	Edit Distance	☐ Solved ☐ Review
3	Regular Expression Matching	☐ Solved ☐ Review
4	Best Time to Buy and Sell Stock IV	☐ Solved ☐ Review
5	Burst Balloons	☐ Solved ☐ Review
6	Split Array Largest Sum	☐ Solved ☐ Review
7	Russian Doll Envelopes	☐ Solved ☐ Review
8	Maximum Profit in Job Scheduling	☐ Solved ☐ Review

DAY 21 — Heaps & Priority Queues *Medium*

#	Problem	Status
1	Kth Largest Element in an Array	☐ Solved ☐ Review
2	Top K Frequent Words	☐ Solved ☐ Review
3	K Closest Points to Origin	☐ Solved ☐ Review
4	Task Scheduler	☐ Solved ☐ Review
5	Reorganize String	☐ Solved ☐ Review

6	Find K Pairs with Smallest Sums	☐ Solved ☐ Review
7	Kth Smallest Element in a Sorted Matrix	☐ Solved ☐ Review
8	Merge k Sorted Lists	☐ Solved ☐ Review

DAY 22 — Tries & Backtracking *Medium*

#	Problem	Status
1	Implement Trie (Prefix Tree)	☐ Solved ☐ Review
2	Design Add and Search Words Data Structure	☐ Solved ☐ Review
3	Word Break	☐ Solved ☐ Review
4	Permutations	☐ Solved ☐ Review
5	Subsets	☐ Solved ☐ Review
6	Combination Sum	☐ Solved ☐ Review
7	Letter Combinations of a Phone Number	☐ Solved ☐ Review
8	Generate Parentheses	☐ Solved ☐ Review

DAY 23 — Backtracking – Advanced *Medium–Hard*

#	Problem	Status
1	N-Queens	☐ Solved ☐ Review
2	Sudoku Solver	☐ Solved ☐ Review
3	Word Search II	☐ Solved ☐ Review
4	Palindrome Partitioning	☐ Solved ☐ Review
5	Combination Sum II	☐ Solved ☐ Review
6	Permutations II	☐ Solved ☐ Review
7	Restore IP Addresses	☐ Solved ☐ Review
8	Subsets II	☐ Solved ☐ Review

DAY 24 — Math, Bit Manipulation & Greedy *Easy–Medium*

#	Problem	Status
1	Single Number	☐ Solved ☐ Review
2	Number of 1 Bits	☐ Solved ☐ Review
3	Counting Bits	☐ Solved ☐ Review
4	Reverse Bits	☐ Solved ☐ Review
5	Power of Two	☐ Solved ☐ Review
6	Gas Station	☐ Solved ☐ Review
7	Partition Labels	☐ Solved ☐ Review
8	Jump Game II	☐ Solved ☐ Review

DAY 25 — Mock Interview Day – Mixed Hard Problems *Hard*

#	Problem	Status
1	Median of Two Sorted Arrays	☐ Solved ☐ Review
2	Trapping Rain Water	☐ Solved ☐ Review
3	Find Median from Data Stream	☐ Solved ☐ Review
4	Word Ladder II	☐ Solved ☐ Review
5	Largest Rectangle in Histogram	☐ Solved ☐ Review
6	Serialize and Deserialize Binary Tree	☐ Solved ☐ Review
7	Maximum Profit in Job Scheduling	☐ Solved ☐ Review
8	LFU Cache	☐ Solved ☐ Review

Progress Tracker

Week	Topics	Started	Completed	%
Week 1 (Days 1–5)	Arrays, Strings	___/___	___/___	___%
Week 2 (Days 6–10)	Linked Lists, Stacks, Trees	___/___	___/___	___%
Week 3 (Days 11–15)	Trees, BST, Binary Search, Hashing, Graphs	___/___	___/___	___%
Week 4 (Days 16–20)	Graphs, DP Basics, DP Advanced	___/___	___/___	___%
Week 5 (Days 21–25)	Heaps, Tries, Backtracking, Bits, Mock	___/___	___/___	___%

Daily Habit Checklist

- ☐ Solve all 8 problems
- ☐ Analyze time & space complexity
- ☐ Review any unsolved problems next morning
- ☐ Take brief notes on new patterns learned
- ☐ Attempt at least 1 problem from previous days